NATO field-trial SERS dataset

Progress update, results and paper options

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Meeting with Li-Lin · 21 October 2026

Since we last spoke (14 September)

Already reported: the classical machine-learning baseline. Extra Trees identified the chemical on an instrument it had never seen with 68.5% balanced accuracy (chance: 33%).

New since then

1. Compact neural networks (CNNs), including versions trained to ignore instrument differences.
2. Three preprocessing pipelines — minimal, smoothing, baseline correction — tested on every model.
3. Your question from 1 September: do the substrates detect the analyte independently of the instrument?

For discussion

- What paper this dataset supports, and where to submit it.
- What a new dataset would need for the original neural-network goal.

What we set out to do

Original idea

Train a disentangled variational autoencoder (VAE) to split each spectrum into a chemical part and a substrate/instrument part.

Why this dataset cannot support it

- No pure-analyte reference spectra, so a "chemical part" cannot be checked.
- Chemicals, substrates and instruments are not crossed: 374 of 2,760 sample $\times$ substrate $\times$ instrument combinations were measured.
- Only 69 independently prepared samples.

Current focus

Your clarification: the trial asked whether the SERS substrates detect the analyte *independently of the instrument*.

Main question

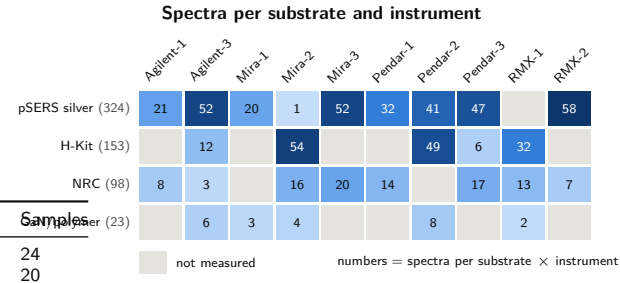
Can a model identify the chemical on an instrument — and on samples — that it never saw during training?

Follow-up questions

When does preprocessing help? Do neural networks add anything over classical methods? Which substrates hold up across instruments?

The dataset

1,172 logged recordings
721 with a named SERS substrate
626 pass numeric quality checks
615 traceable to a sample and target
598 unique spectra
69 independent physical samples

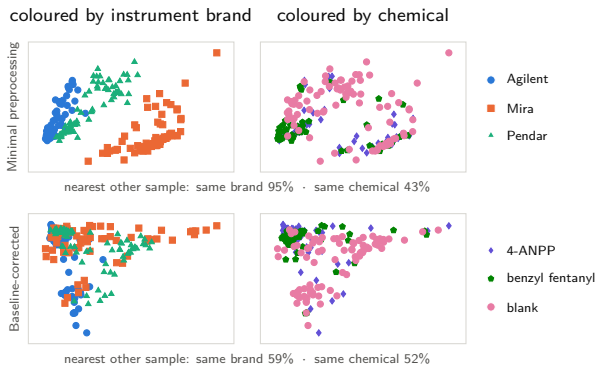


Station	Chemicals	Samples
CWA	4-nitrophenol, ethanol, ethyl paraoxon	24
Pills	4-ANPP, benzyl fentanyl, blank	20
Surfaces	4-ANPP, acetaminophen, benzyl fentanyl	25

Two reduced sets check the results: 500 spectra (quality-flagged removed) and 575 (Mira-1 removed; it failed its system check).

The 598 spectra are repeat views of 69 samples. Every test keeps all views of one sample on the same side of the train/test split.

Instrument differences dominate the raw spectra



Pills station: principal-component map of 208 spectra (one point per spectrum).

Minimal preprocessing: a spectrum's nearest neighbour from another sample has the same instrument brand 95% of the time, but the same chemical only 43%. Both would be about 1 in 3 by chance.

Baseline correction weakens the brand grouping (59% brand, 52% chemical). At Surfaces it reverses: 42% brand, 74% chemical.

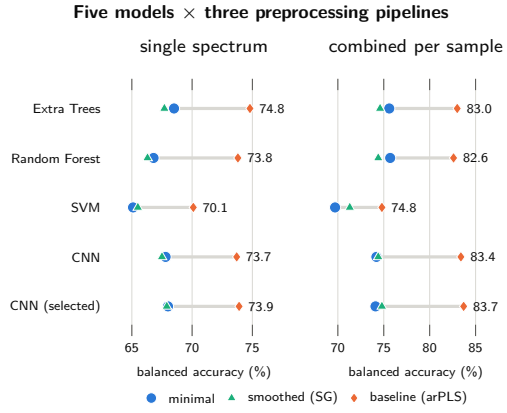
Removing the background moves spectra from grouping by instrument toward grouping by chemical. It does not remove the instrument effect completely.

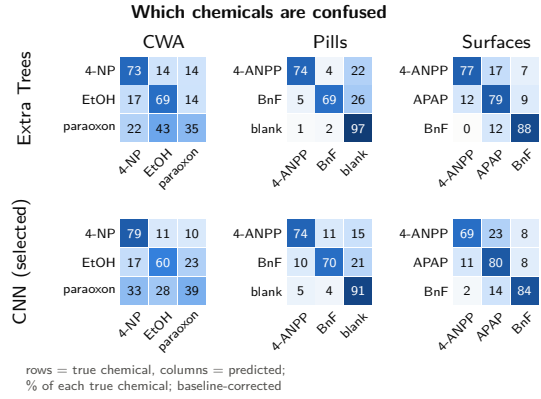
Seven questions and where each stands

	Question	Status
✓	1. Identify the chemical on an unseen instrument?	65–68% minimal; 70–75% baseline-corrected (chance 33%)
✓	2. Which single preprocessing pipeline helps most?	Baseline correction: +5 to +7 points; smoothing: no gain
~	3. Should preprocessing adapt to the instrument?	By brand: not testable here. By spectrum quality: possible at CWA only; not yet run
✓	4. Do neural networks beat classical methods?	No clear difference
○	5. How much do 1–2 calibration samples from a new instrument help?	Not yet run; feasible for 1–2 samples per chemical
○	6. Are confidence scores reliable; can unknowns be flagged?	Not yet run; CNN confidence is poorly calibrated
~	7. Do substrates work independently of the instrument?	Partly: depends on the instrument; Mira units are the weakest

✓ answered ~ partly answered ○ not yet run. Balanced accuracy averages the hit rate over chemicals, so 33% is chance for three chemicals.

Results: identifying chemicals on an unseen instrument





Baseline correction helps every model; tree models and CNNs perform alike. With minimal preprocessing, Extra Trees calls 42–45% of Pills drug spectra "blank"; correction roughly halves this. Ethyl paraoxon is the hardest chemical (35–39% correct).

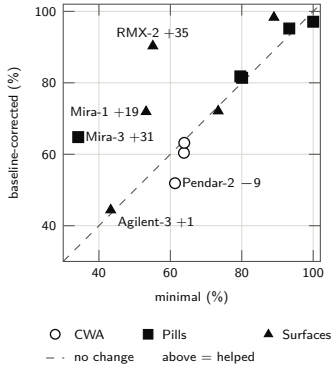
"Combined per sample" averages the model's probabilities over a sample's repeat spectra (spectra are not averaged). 4-NP = 4-nitrophenol, EtOH = ethanol, BnF = benzyl fentanyl, APAP = acetaminophen.

Results: where it works and where it does not

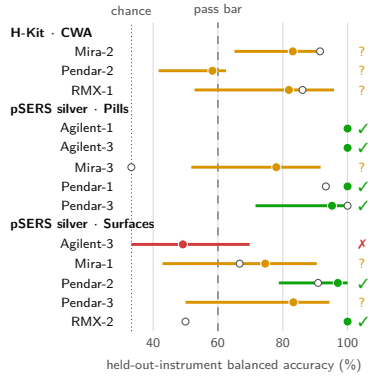
Per chemical, all models (% correct)

		Extra Trees	Rand. Forest	SVM	CNN	CNN (sel.)
CWA	4-nitrophenol	73	73	69	78	79
	ethanol	69	68	67	60	60
	ethyl paraoxon	35	28	37	39	39
Pills	4-ANPP	74	73	72	74	74
	benzyl fentanyl	69	69	66	70	70
	blank	97	93	97	91	91
Surfaces	4-ANPP	77	71	65	68	69
	acetaminophen	79	80	73	80	80
	benzyl fentanyl	88	88	75	83	84

Per instrument (Extra Trees)



Per substrate, held-out instrument



Correction rescues background-heavy instruments (Mira-3 +31, RMX-2 +35 points) but lowers every CWA instrument (by 1–9 points). pSERS silver passes 6 of 10 instrument tests. The field log recorded detection for 75% of pSERS and 23% of H-Kit measurements.

Substrate panel (Extra Trees, chosen after the results): dot and bar = corrected estimate and 95% interval; open circle = minimal. Pass: lower bound $\geq$60% and $\leq$10-point loss. The pre-planned test found no substrate passed; NRC and GaN/polymer had too few samples.

Paper 1: what this dataset supports now

Working title: Chemical identification on unseen handheld Raman instruments: a leakage-controlled benchmark on a multi-instrument SERS field trial

Contributions

- Curated field-trial dataset with its coverage matrix.
- Strict test: unseen instrument *and* unseen samples.
- Tree models and compact CNNs perform alike.
- Baseline correction: $+5$ to $+7$ points, but instrument-dependent.
- Substrate performance across instruments is condition-specific.

Candidate journals

Analyst (first choice) · Analytical Chemistry (reach) · Applied Spectroscopy · Analytica Chimica Acta · J. Raman Spectrosc.; Scientific Data if the data can be released

The contribution is the test protocol and a rare dataset, not a new method. Classical $\approx$ CNN agrees with recent Raman benchmarks (Sineesh & Kamsali, *Digital Discovery* 2026). Checked 20 October 2026 against Crossref; full list in backup.

Closest prior work

- Fornasaro et al. and Guo et al., *Anal. Chem.* 2020: spectra differ across labs and devices — variability measured, identification not benchmarked.
- Mahmud et al., *Spectrochim. Acta A* 2026: maps 4 portable SERS instruments to a lab instrument — bacteria, one substrate.
- Lai et al., *Anal. Chem.* 2025; She et al., *Anal. Chem.* 2026: transfer models between devices — need data from the new instrument.
- Mirsafavi et al., *Analyst* 2020; Wilson et al., *Sens. Actuators B* 2021: SERS of fentanyl and 4-ANPP — one instrument.

Gap: no study tests an unseen handheld instrument with sample-grouped splits across several substrates and vendors.

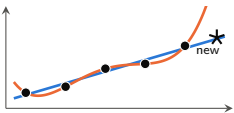
What a proper dataset needs

Independent samples per chemical × substrate

	pSERS	H-Kit	NRC	GaN
4-nitrophenol		8	5	4
ethanol		9	4	
ethyl paraoxon		7	3	
4-ANPP	11		7	
benzyl fentanyl	15		7	1
acetaminophen	11	1	4	
blank	8		8	

never measured together

Why sample count matters (schematic)



1. Fill the matrix. A model separates two effects only if they vary independently. Here H-Kit met only CWA chemicals and pSERS only Pills/Surfaces chemicals. Needed: every chemical on every substrate and instrument, several independent samples per cell.

2. Ground truth for each effect: pure-analyte Raman references, substrate-only blanks, instrument calibration standards; otherwise a model's "chemical part" cannot be checked.

3. Enough independent samples. Our CNN has 208,691 adjustable parameters for 69 samples (≈3,000 per sample). Extra parameters let a model fit the training data exactly yet miss new samples (left, bottom).

Illustrative design: (6 analytes × 3 concentrations + 1 blank) × 4 substrates × 5 preparations = 380 independent samples (5.5× today), each read on 4 instruments, including ≥3 units of one brand.

Theory agrees: without labelled factors, disentanglement cannot be learned (Locatello et al., ICML 2019); correlated factors stay entangled (Träuble et al., ICML 2021). Schematic uses five invented points; no exact "samples > parameters" rule exists.

Paper 2: the original VAE goal, with a designed dataset

Working title: Separating analyte, substrate and instrument contributions in SERS spectra with structured variational autoencoders

How it would be tested

- Recover each analyte's reference spectrum from its SERS spectra.
- Predict chemical × substrate × instrument combinations deliberately left out of training.
- Compare against the Paper 1 benchmark methods.

Why Paper 1 comes first

Paper 1 shows the instrument effect dominates, and why separating it requires a crossed design.

Closest prior work

- Fornasaro et al., *Anal. Chem.* 2020: 15 labs, 6 SERS methods — but one analyte; authors caution against extrapolating to other substrates.
- Mahmud et al., *Spectrochim. Acta A* 2026: 20 analytes, 5 instruments — but one substrate.
- Abram & McCloskey, *Biomolecules* 2023: a VAE separates instrument type — in mass spectrometry.
- Lotfollahi et al., *Mol. Syst. Biol.* 2023: predicts unseen drug × cell-type combinations — a template from biology.

Gap: no crossed analyte × substrate × instrument SERS dataset with disentangled models was found.

Candidate journals

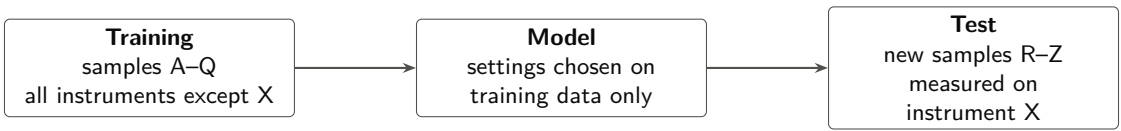
Analytical Chemistry (ACS) · Digital Discovery (RSC) · Machine Learning: Science and Technology (IOP) · Nature Communications (stretch) · plus a Scientific Data descriptor for the dataset

Literature checked 20 October 2026 against Crossref and publisher/proceedings pages; repeat in Scopus before submission. Full list in backup.

For discussion

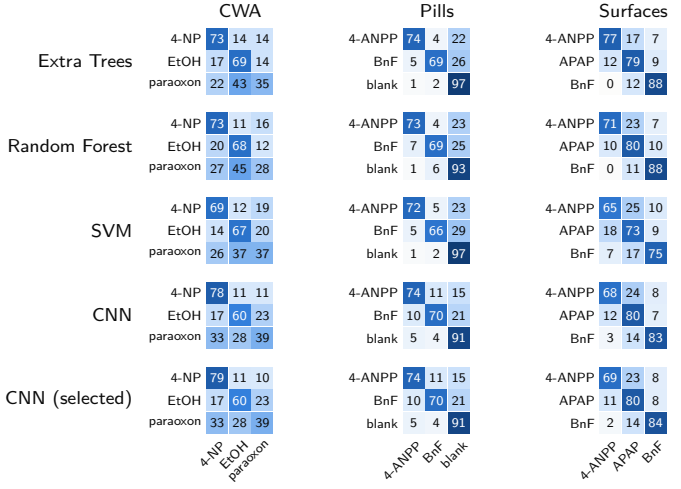
1. **Publication.** Can results from the NATO trial data be published? Who must approve, who are co-authors, and is there a citable trial report (NATO SET-292, Aberdeen, March 2024)?
2. **Route.** Paper 1 now with this dataset; Paper 2 requires new measurements.
3. **New data.** Which analytes, substrates and instruments could NRC measure? Are $\geq$3 units of one brand available? Can one SERS chip be read on several instruments before it degrades?
4. **Chemistry.** Why might the Mira units fail without baseline correction (Mira-1 also failed its system check)? Was ethyl paraoxon deposited from an ethanol solution?
5. **Next experiments.** 1–2-sample calibration on a new instrument; robustness to added noise and background; other normalizations. Which matter most for the paper?

Backup: how the test was kept fair



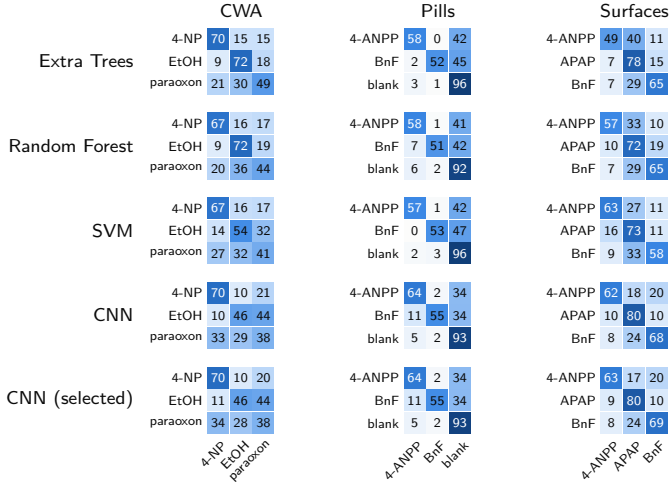
- Every spectrum of one physical sample stays on one side of the split (four folds, repeated five times).
- The held-out instrument contributes nothing to training, preprocessing statistics, model selection, calibration or stopping.
- 13 station–instrument combinations have enough samples for this test (CWA: 3, Pills: 5, Surfaces: 5).
- Scores use the same method as the 14 September report, so the 68.5% Extra Trees figure is unchanged.

Backup: all confusion matrices, baseline-corrected



Rows = true chemical, columns = predicted; % of each true chemical.

Backup: all confusion matrices, minimal preprocessing



Rows = true chemical, columns = predicted; % of each true chemical.

Backup: accuracy per instrument (%), minimal → corrected

Station	Instrument	Extra Trees	Random Forest	SVM	CNN	CNN (selected)
CWA	Mira-2	64 → 60	58 → 58	50 → 60	50 → 60	50 → 60
CWA	Pendar-2	61 → 52	60 → 48	52 → 50	49 → 59	49 → 59
CWA	RMX-1	64 → 63	63 → 63	60 → 63	54 → 60	55 → 60
Pills	Agilent-1	80 → 81	79 → 83	81 → 81	80 → 79	80 → 79
Pills	Agilent-3	93 → 95	88 → 93	96 → 92	93 → 87	93 → 87
Pills	Mira-3	34 → 65	34 → 60	33 → 64	38 → 69	38 → 69
Pills	Pendar-1	80 → 82	76 → 83	78 → 80	83 → 80	83 → 80
Pills	Pendar-3	100 → 97	100 → 97	100 → 95	99 → 95	99 → 95
Surfaces	Agilent-3	43 → 44	35 → 46	47 → 46	46 → 55	46 → 53
Surfaces	Mira-1	53 → 72	59 → 73	50 → 63	56 → 66	55 → 68
Surfaces	Pendar-2	89 → 98	86 → 98	55 → 64	93 → 98	95 → 98
Surfaces	Pendar-3	73 → 72	67 → 70	69 → 64	72 → 70	72 → 70
Surfaces	RMX-2	55 → 90	62 → 87	73 → 88	67 → 81	68 → 83

Backup: remaining preprocessing experiments

Experiment	Doable?	Reason
Other normalizations (SNV, vector, area, derivative)	Yes	Classical models only, ≈42k fits; hours of computing
Wider spectral range (400–1,849 cm^{-1})	Yes	≈63k fits; hours
Robustness to added noise, background, spikes, shifts	Yes	Reuses trained models; mostly prediction
Reduced datasets (500 / 575 spectra)	Yes	Classical-only version is quick; with CNNs ≈539k fits
Preprocessing chosen from spectrum quality	Partly	Supported in 54 of 260 tests, all at CWA; full search ≈1.6M fits
Preprocessing chosen per instrument brand	No	Instruments were tied to stations: within a station, holding one out leaves $\leq$1 unit of that brand to learn from

Backup: glossary

Balanced accuracy	Hit rate per chemical, averaged over chemicals. Chance is 33% for three chemicals.
Held-out instrument	An instrument whose spectra are used only for testing.
Independent sample	A separately prepared physical sample; repeat spectra of it are not independent.
Parameter	One adjustable number a model learns from data (a weight).
Extra Trees / Random Forest	Many decision trees voting; Extra Trees chooses its splits more randomly.
SVM	Support-vector machine: finds the boundary with the widest margin between chemicals.
CNN	Convolutional neural network: learns local spectral patterns such as peak shapes.
arPLS	Asymmetrically reweighted penalized least squares: estimates and subtracts a smooth background.
SG smoothing	Savitzky–Golay filter: local polynomial smoothing (11 points, degree 3).
Confusion matrix	Table of true versus predicted chemical.

Backup: references for Paper 1

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Backup: references for Paper 2

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